

INSA Rouen Normandie – LITIS Laboratory

Master’s Research Internship

When Is Graph Structure Actually Useful?

Edge reconstructibility and downstream performance of MLPs vs GNNs

Location	INSA Rouen Normandie, Madrillet campus, LITIS Laboratory, Rouen, France
Start date	Flexible, from January 2027
Duration	5 to 6 months, full time
Level	Second-year Master’s student or final-year engineering student
Field	Machine learning, graph machine learning, experimental evaluation
Project	FAMOUS, funded by the French National Research Agency (ANR-23-CE23-0019)
Contact	Benoit Gaüzère – benoit.gauzere@insa-rouen.fr

Background

Graph Neural Networks (GNNs) are designed to jointly exploit node attributes and the relations encoded by graph edges. Their use relies on the assumption that graph structure provides useful information beyond node features. However, several recent studies [2, 3, 5] have shown that models that ignore edges, such as multilayer perceptrons (MLPs), can achieve comparable performance on some common benchmarks.

Within the FAMOUS project, we have developed an *edge reconstructibility* diagnostic [1]. The principle is to train a model to predict whether an edge exists using only the attributes of its incident nodes, without providing the adjacency matrix, node degrees, or neighbourhood information. A structure that can easily be reconstructed may be largely redundant with the available attributes. Conversely, a structure that is difficult to reconstruct potentially contains additional information.

Our initial results show that reconstructibility strongly depends on the graph construction mechanism and on the difficulty of the non-edges used for evaluation [4]. More importantly, they raise an open question: **can this diagnostic explain when a GNN actually improves performance on the target task?**

Scientific Objective

The main objective of this internship is to investigate the relationship between the reconstructibility profile of a graph and the gain achieved by a GNN over a model that uses node attributes only:

$$\Delta_{\text{GNN}} = \text{Perf}_{\text{GNN}} - \text{Perf}_{\text{MLP}}.$$

The intern will determine whether specific forms of reconstructibility, particularly under challenging negative samples, can explain or predict Δ_{GNN} . The work will also characterise counterexamples: graph structure that cannot be reconstructed but is irrelevant to the target task, or reconstructible structure that nevertheless provides an effective inductive bias for a GNN.

Main Tasks

The internship will be organised into four work packages.

1. **Consolidate the reconstructibility diagnostic.** Properly extend the protocol to undirected graphs; add metrics such as ROC-AUC and Average Precision; implement simple baselines, including Euclidean distance, cosine similarity, logistic regression.
2. **Extend the experimental study.** Add datasets spanning several families and covering datasets used in the literature (TU Dataset, Pytorch Geometric Dataset, Open Graph Benchmark). For datasets containing spatial coordinates, conduct ablation studies to quantify their specific contribution to reconstructibility.
3. **Investigate global reconstructibility.** Implement a model that uses the complete set of node attributes X while remaining strictly unaware of the graph structure. Architectures such as DeepSets or Set Transformers may be explored to determine whether some edges can only be reconstructed by placing node pairs in their global feature context.
4. **Relate the diagnostic to downstream tasks.** Compare feature-only models and standard GNNs on the native tasks associated with each dataset under fair experimental conditions by reproducing experiments and extracting results from published papers. Statistically analyse the relationship between reconstructibility profiles and the GNN/MLP performance gap, while accounting for data family, homophily, graph density, task type, ...

The **fourth work package is the scientific core of the internship**. The first three provide the methodology, data, and model variants required for this analysis.

Proposed Methodology

As already published, the diagnostic will be evaluated using several negative sampling strategies: uniform sampling, proximity in feature space, local structural scores such as Resource Allocation, global proximity based on Personalized PageRank, and local *edge-flip* perturbations. The reconstruction model will never receive information derived from the adjacency matrix; graph structure may only be used to construct challenging negative examples.

Downstream experiments will include at least one carefully tuned MLP baseline and standard GNN architectures appropriate for each task, such as GCN, GraphSAGE, GAT, or GIN. Models will be evaluated on identical splits, with consistent hyperparameter budgets, multiple random initialisations, and confidence intervals.

The expected outcome is a reconstructibility profile for each dataset and an analysis of its relationship with Δ_{GNN} , rather than a ranking of link-prediction models.

Expected Outcomes

- a reproducible implementation of the diagnostic for directed and undirected graphs;
- a documented collection of baselines and a multi-dataset benchmark;
- a comparison of local and global reconstruction from node attributes;
- a statistical analysis of the relationship between reconstructibility and downstream performance;
- we aim to publish a scientific paper extending the current conference paper.

Candidate Profile

We are looking for a second-year Master’s student or final-year engineering student with:

- a solid background in machine learning and deep learning;
- a strong interest in graphs, GNNs, and experimental evaluation;

- good Python programming skills; experience with PyTorch and, ideally, PyTorch Geometric would be appreciated;
- a rigorous approach to experimentation, statistical analysis, and reproducibility;
- the ability to read scientific papers and communicate results in English;
- scientific curiosity, autonomy, and a critical approach to empirical results.

Previous experience in graph machine learning is an advantage, but is not required for applicants with strong foundations in machine learning and programming.

Research Environment and Supervision

The internship will take place at the LITIS Laboratory within the App team, on the Madrillet campus of INSA Rouen Normandie. It will be part of the FAMOUS ANR research project and will build on the code, initial results, and scientific environment associated with the ongoing manuscript on edge reconstructibility.

The internship will be primarily supervised by **Benoît Gaüzère** and **Alexis Imbert**, PhD student and first author of the conference paper. Scientific discussions may involve other researchers contributing to the project and the initial manuscript. The intern will participate in team meetings and will have access to the required computing resources.

Indicative Schedule

Period	Main activities
January 2027	Onboarding, reproduction of existing results, and finalisation of the protocol and dataset selection.
February	Undirected graphs, simple baselines, evaluation metrics, and experiment automation.
March	Global model over X and SOTA on GNN vs MLP benchmarks.
April	Launch of the downstream benchmark.
May	MLP/GNN comparison and statistical analyses.
June	Final experiments, Master's thesis, oral presentation, and preparation of a manuscript.

Application

Applications should include:

- a CV;
- a cover letter or short message explaining the applicant's interest in the topic;
- contact details of people who already interact professionally or during your studies with you;
- if available, an example of a relevant project, code repository, or scientific report. Github profile or similar is particularly appreciated;

Please send applications to benoit.gauzere@insa-rouen.fr with the subject line *Application – FAMOUS MSc Internship 2027*. Applications will be reviewed on a rolling basis until the position is filled.

Selected References

References

- [1] A. Imbert, B. Gauzere, S. Takerkart, G. Auzias, and P. Honeine. *On the Predictability of Graph Structure from Node Features*. S+SSPR 2026, Bern.
- [2] S. Luan et al. *When Do We Need Graph Neural Networks for Node Classification?* Complex Networks & Their Applications, 2024.
- [3] I. Katsman, E. Lou, and A. Gilbert. *Revisiting the Necessity of Graph Learning and Common Graph Benchmarks*. arXiv:2412.06173, 2024.
- [4] J. Li et al. *Evaluating Graph Neural Networks for Link Prediction: Current Pitfalls and New Benchmarking*. NeurIPS, 2023.
- [5] M. Bechler-Speicher et al. *Graph Neural Networks Use Graphs When They Shouldn't*. ICML, 2024.